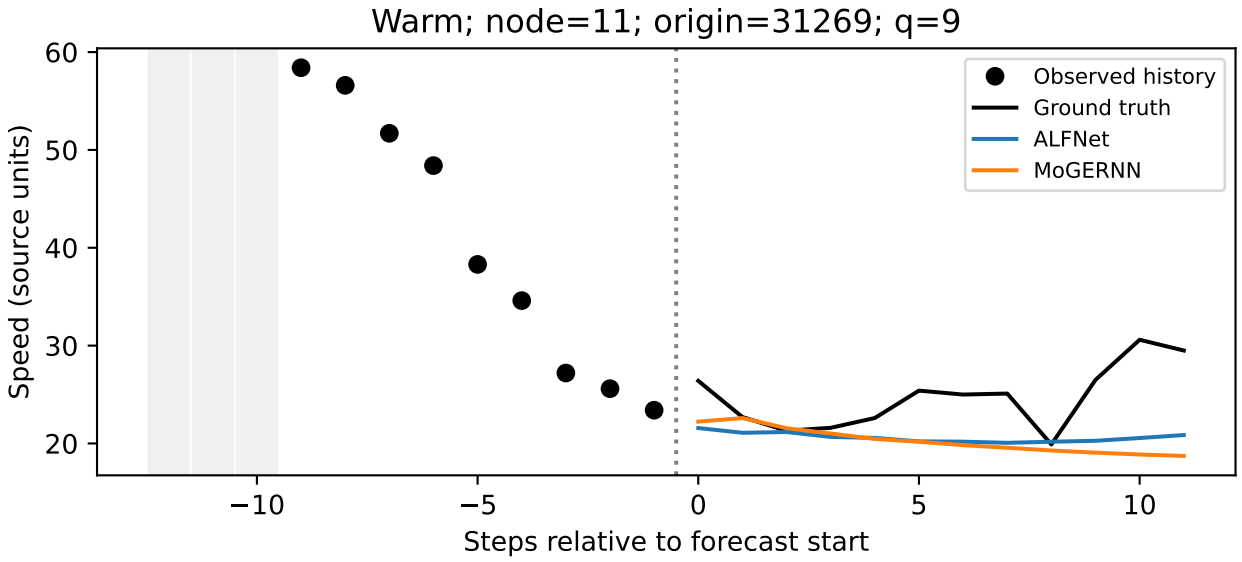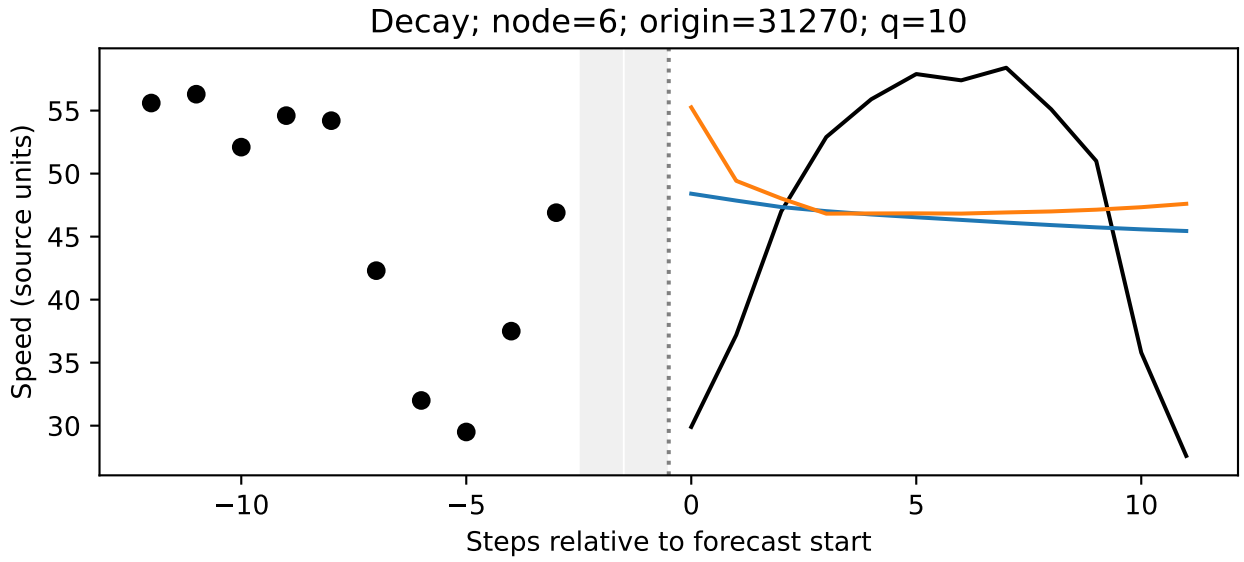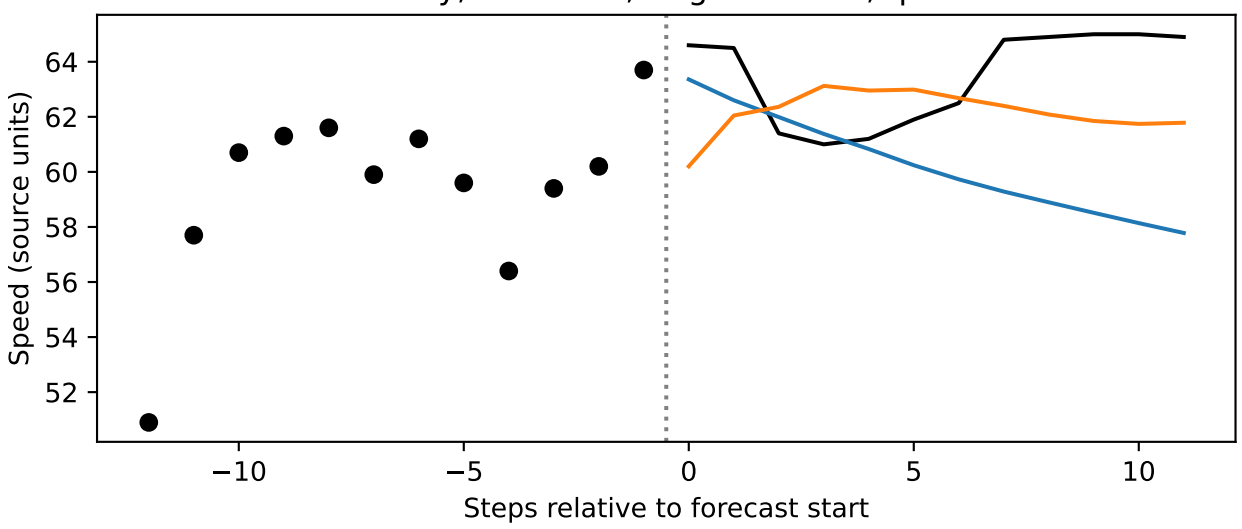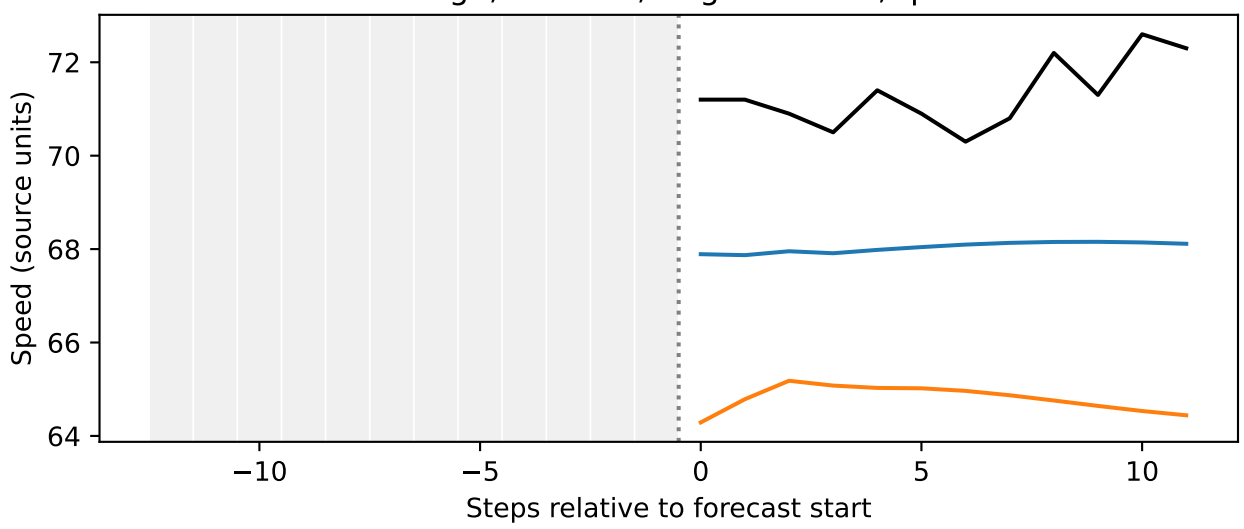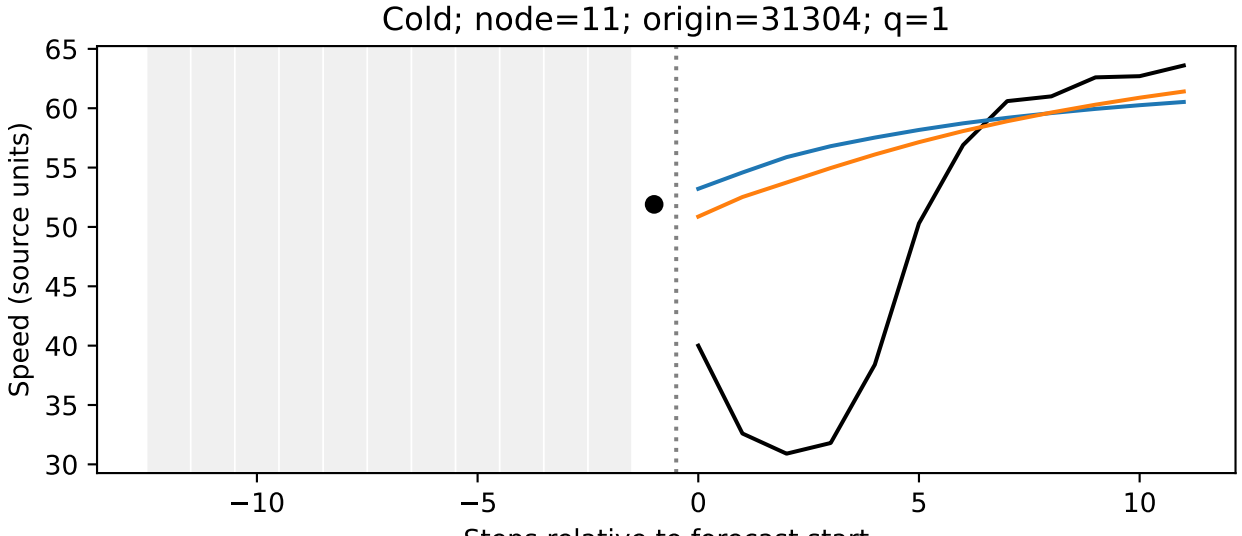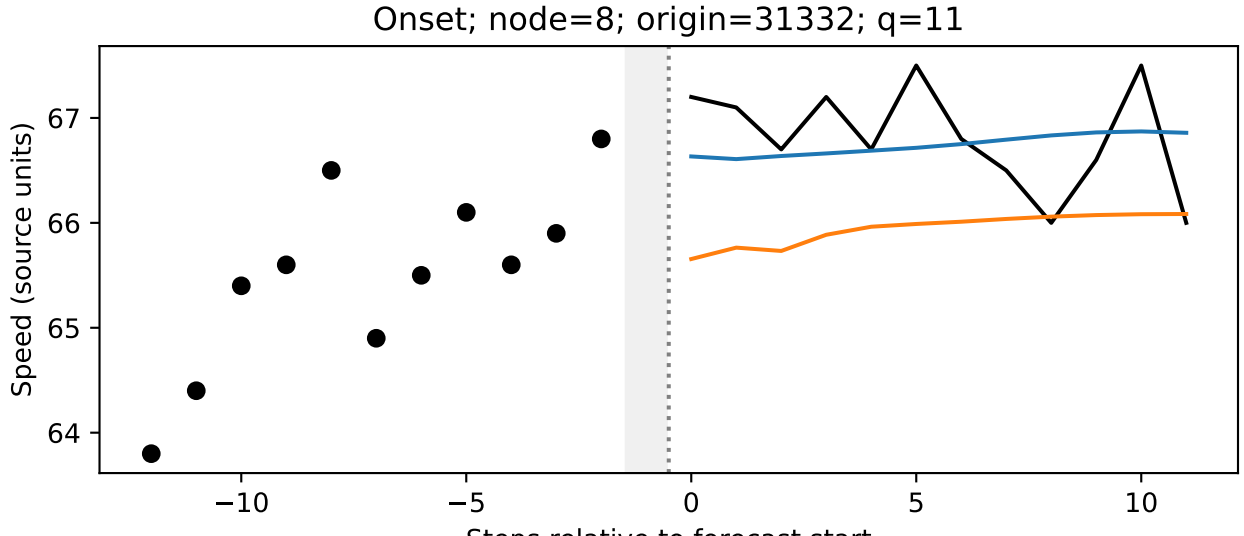PEMSBAY deterministic validation examples; not causal mechanism evidence